

COMP3013 2024 Fall

Assignment 3

For the submission, please pack all files and convert them into **A SINGLE PDF FILE**. Rename the PDF file as COMP3013_24F_A3_XXXX, where XXXX is your student ID.

The schema of a database for public transportation companies is as follows. Keys are underlined.

- *company* = (*cID*, *cname*, *address*, *phone*)
- *route* = (*rID*, *departure*, *arrival*, *cID*)
 // *cID* is a foreign key to *company.cID*.
- *vehicle* = (*plateNum*, *model*, *capacity*, *manufacturer*, *cID*)
 // *cID* is a foreign key to *company.cID*.
- *driver* = (*dID*, *name*, *gender*, *age*, *cID*)
 // *cID* is a foreign key to *company.cID*.
- *serve* = (*rID*, *plateNum*)
 // *rID* is a foreign key to *route.rID*.
 // *plateNum* is a foreign key to *vehicle.plateNum*.
- *drive* = (*plateNum*, *dID*)
 // *dID* is a foreign key to *driver.dID*.
 // *plateNum* is a foreign key to *vehicle.plateNum*.

Q1. Write a query for each following question. (10pt for each)

- a) Find the ID of routes which are not served by any vehicle. You must use subqueries.
- b) Find the name of drivers who have served the route 69 (*rID*). You must use subqueries.
- c) Find the name of drivers who have driven all vehicles.
- d) Find the plate number of vehicles which have served all route operated by “Xinhe” (company name).
- e) Implement constraints to guarantee the gender of a driver is either “Male” or “Female” and the age is from 20 to 60.

```
-- a) Find the ID of routes which are not served by any vehicle.
You must use subqueries.
SELECT rID
FROM route
WHERE rID NOT IN (
    SELECT rID
    FROM serve
);

-- b) Find the name of drivers who have served the route 69 (rID).
You must use subqueries.
SELECT name
FROM driver
WHERE dID IN (
    SELECT dID
    FROM driver
```

```

        WHERE plateNum IN (
            SELECT plateNum
            FROM serve
            WHERE rID = 69
        )
    );

-- c) Find the name of drivers who have driven all vehicles.
SELECT name
FROM driver
WHERE dID IN (
    SELECT dID
    FROM drive
    GROUP BY dID
    HAVING COUNT(DISTINCT plateNum) = (
        SELECT COUNT(*)
        FROM vehicle
    )
);

-- d) Find the plate number of vehicles which have served all
route operated by "Xinhe" (company name).
SELECT plateNum
FROM vehicle
WHERE plateNum IN (
    SELECT plateNum
    FROM route
    WHERE rID IN (
        SELECT rID
        FROM route
        WHERE cID = (
            SELECT cID
            FROM company
            WHERE cname = "Xinhe"
        )
    )
    GROUP BY plateNum
    HAVING COUNT(DISTINCT rID) = (
        SELECT COUNT(*)
        FROM route
        WHERE cID = (
            SELECT cID
            FROM company
            WHERE cname = 'Xinhe'
        )
    )
);

-- e) Implement constraints to guarantee the gender of a driver
is either "Male" or "Female" and the age is from 20 to 60.
ALTER TABLE driver
ADD CONSTRAINT chk_gender CHECK (gender IN ('Male', 'Female')),
ADD CONSTRAINT chk_age CHECK (
    age BETWEEN 20 AND 60

```

);

Q2. Given an instance of a relational schema $R = \{A, B, C\}$ and a list of functional dependencies.

A	B	C
1	2	2
1	3	2
1	4	2
2	5	2

Decide whether the functional dependency is satisfied by the instance. (10 pt)

- | | | |
|-----------------------|-----------------------|-----------------------|
| a) $A \rightarrow B$ | b) $A \rightarrow C$ | c) $B \rightarrow A$ |
| d) $B \rightarrow C$ | e) $C \rightarrow A$ | f) $C \rightarrow B$ |
| g) $AB \rightarrow C$ | h) $AC \rightarrow B$ | i) $BC \rightarrow A$ |

- | | |
|-----------------------|--|
| a) $A \rightarrow B$ | when $A = 1$, $B = \{2, 3, 4\}$, so (a) is not satisfied . |
| b) $A \rightarrow C$ | when $A = 1$, $C = 2$; when $A = 2$, $C = 2$; so (b) is satisfied . |
| c) $B \rightarrow A$ | when $B = 2, 3, 4$, $A = 1$; when $B = 5$, $A = 2$; so (c) is satisfied . |
| d) $B \rightarrow C$ | $B = 2, 3, 4, 5$, $C = 2$; so (d) is satisfied . |
| e) $C \rightarrow A$ | $C = 2$, $A = \{1, 2\}$, so (e) is not satisfied . |
| f) $C \rightarrow B$ | $C = 2$, $B = \{2, 3, 4, 5\}$, so (f) is not satisfied . |
| g) $AB \rightarrow C$ | $AB = \{(1, 2), (1, 3), (1, 4), (2, 5)\}$, $C = 2$, so (g) is satisfied . |
| h) $AC \rightarrow B$ | when $AC = (1, 2)$, $B = \{2, 3, 4, 5\}$, so (h) is not satisfied . |
| i) $BC \rightarrow A$ | $BC = \{(2, 2), (3, 2), (4, 2), (5, 2)\}$, $A = \{1, 2\}$, (i) is satisfied . |

Q3. Given a relational schema and a set of functional dependencies

- $R = \{A, B, C, D, E\}$
- $F = \{AB \rightarrow CD, BC \rightarrow DE, CD \rightarrow E, DE \rightarrow A\}$

- Find all candidate keys of R . (6 pt)
- Decompose R into BCNF. Show the steps. (15 pt)
- Does the BCNF decomposition in part b) preserve all functional dependencies? (4 pt)
- Decompose R into 3NF. Show the steps. (15 pt)

- a)
- $$AB^+ = \{A, B, C, D, E\}, BC^+ = \{A, B, C, D, E\}, BDE^+ = \{A, B, C, D, E\}$$

Candidate keys are AB, BC, and BDE.

- b)

Firstly, we know CD and DE is not super key, which violates BCNF.

Thus we need to decompose R .

For the dependency $CD \rightarrow E$, we decompose R into:

$$R_1 = \{C, D, E\}, F_1 = \{CD \rightarrow E\}$$

$$R_2 = \{A, B, C, D\}, F_2 = \{AB \rightarrow CD, BC \rightarrow DE, DE \rightarrow A\}$$

For the dependency $DE \rightarrow A$, we decompose R_2 into:

$$R_3 = \{D, E, A\}, F_3 = \{DE \rightarrow A\}$$

$$R_4 = \{B, C, D\}, F_4 = \{AB \rightarrow CD, BC \rightarrow DE\}$$

Thus, the BCNF decomposition results in the following relations:

$$R_1 = \{C, D, E\}$$

$$R_3 = \{A, D, E\}$$

$$R_4 = \{B, C, D\}$$

c)

1. $AB \rightarrow CD$

$AB \rightarrow C$ lost, $BC \rightarrow D$ in R_4

2. $BC \rightarrow DE$

$BC \rightarrow D$ in R_4 , $CD \rightarrow E$ in R_1

3. $CD \rightarrow E$ in R_1

4. $DE \rightarrow A$ in R_3

The BCNF decomposition does not preserve all functional dependencies, specifically $AB \rightarrow C$

d)

$$F = \{AB \rightarrow CD, BC \rightarrow DE, CD \rightarrow E, DE \rightarrow A\}$$

1. Convert FDs to Minimal Cover:

$$F_c = \{AB \rightarrow CD, BC \rightarrow DE, CD \rightarrow E, DE \rightarrow A\}$$

D is extraneous in $AB \rightarrow CD$

E is extraneous in $BC \rightarrow DE$

$$F_c = \{AB \rightarrow C, BC \rightarrow D, CD \rightarrow E, DE \rightarrow A\}$$

2. Create a Relation for Each Functional Dependency in the Minimal Cover:

$$R_1 = \{A, B, C\}$$

$$R_2 = \{B, C, D\}$$

$$R_3 = \{C, D, E\}$$

$$R_4 = \{A, D, E\}$$

3. R_1 already contains a candidate key (AB) for R

4. Return $\{ \{A, B, C\}, \{B, C, D\}, \{C, D, E\}, \{A, D, E\} \}$